

Decentralized one stop solution for real estate

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Abstract—Real estate investing has traditionally been a complicated and opaque process, but blockchain technology and tokenization can provide a solution. Investors can gain access to a previously inaccessible asset class by using blockchain-based tokens to represent fractional ownership of real estate. This paper explores the feasibility of a tokenization-based real estate investment project and the pros and cons of using blockchain-based tokens for fractional ownership. The platform, called "Estate-O-Blockchain," will allow users to invest in real estate with digital tokens, creating a new and more accessible form of real estate investment. The platform will provide an easy-to-use interface for buying and selling digital tokens, as well as a secure and transparent platform for tracking investments and real estate ownership. In addition, Estate-O-Blockchain will provide users with access to detailed real estate information, including property history, financial data, and property management information. This project has the potential to revolutionize the real estate investment industry by providing a secure and accessible platform for real estate investment.

Index Terms—Blockchain, Tokenization, Ethereum, Smart Contract, Real-Estate

I. INTRODUCTION

A. Description

Unquestionably one of the most trustworthy investments that individuals have made is real estate. In addition to being a successful investment, real estate also provides a steady stream of income in the form of leases and rentals. Despite the many advantages of real estate investing, one of the biggest negatives is a lack of liquidity. There are fewer investors in the real estate market than there are in the stock market, despite the fact that global real estate investments are almost twice as large as stock market investments. Blockchain technology has the potential to greatly increase market liquidity and transparency while also opening up investment opportunities for private investors. With the capability and adaptability of creating Security Tokens that are backed by genuine assets, real estate may be made liquid. Blockchain tokens, in essence, give a digital representation of total or joint ownership for any entity with a defined value. Blockchain tokens are frequently used for payments and transaction settlement among participants. Additionally, tokens reflect shared ownership of indivisible goods like works of art or music videos. Tokens also make it simpler to transfer ownership of indivisible assets through a blockchain network. Estate-O-BlockChain would help Owners

of houses with liquidity and Buyers with access to invest in Estate with small Investments. Estate properties have been an In-liquid asset class and the most expensive to acquire for normal people. On top of this registration fees and taxes on transfers make matters even worst.

B. Problem Formation

Analysis and Brainstorming about the Real Estate market made us think about different people with different roles and aims with which they join this market. Some are here for investment, some for rental income, some for brokerage and some to flip the property, and many more but the basic problem with all of them is funds. All of them have funds parked in previous investments and cannot be taken out so easily.

C. Motivation

After making a thorough analysis of this market we encountered some known and solved problems while some problems remained unsolved. The gaps we found after the literature survey and analysis of markets are

- **In-Liquid:** As Investment in Estate requires a big Corpus, Count of buyers who can buy reduces drastically. So, the Owner cannot even get the market price for the property. Hence, finding the right buyer is a time-consuming process.
- **Fees and Taxes:** Whenever Property is transferred a hefty amount of registration fees is charged, also the taxes which come with capital gains are a headache.
- **Time Consuming:** As most, Buyer tends to take loans on their property so it makes the transaction process even more lengthy.
- **Estate Handling:** Even after buying Estate it becomes a heavy-duty task to set it up on rent, maintenance, etc.
- **Selling Part Ownership:** It is not feasible with the current system to sell Estates part ownership to raise funds.

D. Proposed Solution

To Counter all of these issues we came up with some solutions each a small part of our project. With these solutions, we hope to solve above described problems.

- **Making Part Ownership possible for Seller.** To achieve these we propose Tokenization on ownership of BlockChain and Blockchain being the owner of all these properties.

- As a Minimum of 50 percent of the Estate is still with the Owner they would take care of handling the property. While they could use raised funds to generate extra income.
- Once the property is verified and Listed on Blockchain it directly becomes a liquid asset with the help of tokenization. However, the Owner can sell his part of the property in the offline market while cooperating with the company handling the network of Blockchain.
- No need to take a loan, Rather a partner with buyers on the network and enjoy EMI-free life. This makes the process less Time-Consuming
- The Part Owners on Blockchain will be out of Tax reach till they could sell their ownership and convert to stablecoins rather than the regulated currency which is not taxed by the government.
- Transfer fees are also less as tokens are kept in smart contracts while just assigned and redeemed from users.

E. Scope Of Project

The main focus of our project is to tokenize the part ownership of listed properties. While each token owner has equal ownership in each and every property which eventually reduces risk by diversification. Given that project provides participants the low risk and moderate returns opportunities on small capital, currently the best security, freedom high-interest rate loan, etc we hope Estate-o-Blockchain has immense potential to grow with the current marketplace. In the future, we hope to map tokens with the buyer's id in a smart contract to remove dependency on the wallet, eliminating grass fees by just maintaining the ledger for the transfer of tokens. The technique used to complete these projects are:

- Smart Contract Blockchain.
- IPFS to store property details and documents safely and return hash.
- Tokenization of non-fungible assets.

II. LITERATURE REVIEW

In this literature survey we will focus on important topics on which our project is based. In this paper, we introduce blockchain technology in the real estate sector. Despite the fact that real estate is an extremely significant element of a country's economy, the current system has various issues, including looking for property, lease agreements, sale and purchase, money transactions, the participation of intermediaries, and so on. Real estate and blockchain can coexist. The land business procedure employs the blockchain model to powerfully set the real estate premise up in many exercises. Blockchain technology has the potential to increase apartment trade activity. Blockchain disintermediation in real estate is a fresh approach for businesses that are continually finding a fraction of these realities. In this article, we will explore blockchain-based technologies employed in the project, such as smart contracts and tokenization. First, let us start with blockchain itself.

A. Blockchain

Blockchain is a secure data storage method that prevents system updates, hacking, and fraud. It is a digital record of transactions that is replicated and distributed across the blockchain's computer network. Transactions are stored in blocks, and each participant's ledger is updated whenever a new transaction occurs on the blockchain. DLT (Distributed Ledger Technology) is a group-managed decentralised database. It records transactions using a cryptographic signature known as a hash, and blockchain is a type of DLT that employs this technology.

B. Smart Contract

Smart contracts are programmes that are executed and stored on a blockchain when certain criteria are met. They are frequently used to automate the execution of an agreement so that all parties have immediate confidence that it will be completed without involving an intermediary or wasting time. They can also automate a workflow by automatically initiating the next activity when certain conditions are met. Ethereum Establishment is an organisation dedicated to the development of decentralised Ethereum virtual machines. EVM enables us to install blockchains and create smart contracts that interact not only with the existing blockchain but also with other blockchains in the network. The Ethereum foundation also created the concept of a DAPP, which stands for decentralised application. They are similar to standard web applications in that they are hosted on a decentralised network rather than a single server. Consider the current challenges in the real estate market, as this will help us understand the impact of blockchain [3].

The article [1] describes a method for utilising blockchain technology in smart contracts, which have proven to be extremely trustworthy and safe and can address the issue that the traditional real estate market is experiencing. The article demonstrated how to use blockchain to keep a real estate rental file online. This paper explains briefly and provides solutions to three traditional problems encountered by real estate management systems based on Centralization Smart Transactions. These issues include transparency, the process of looking for available units and real estate, the accuracy of data, and the financial payment procedure.

The article [2] discusses how blockchain technology can be used to solve a variety of common problems in the land management system. A smart contract is included in addition to BCT to reduce transaction registration time and address concerns about double spending and data manipulation. The proposed method has the potential to enable additional specific cases in LAS, such as sharing property, transferring a portion of property, dividing or combining property, and restricting property trading opportunities, which have not previously been addressed. This is accomplished by developing a programming interface based on two existing interfaces (ERC-20 and ERC-721), neither of which has the required capabilities to address these more specific interfaces.

C. Land registry and rental housing system

Land registration is a time-consuming and inefficient process in India, as it is in many other countries around the world. Furthermore, a number of middlemen are involved in the land registration process. Creating a system that not only accelerates the land registration process but also facilitates the transfer of land ownership from the seller to the new buyer is only possible if a distributed system that stores all transactions during the land acquisition process is created. The [5] study examines the advantages and disadvantages of implementing a blockchain-based system for land ownership transfer.

We can trigger various events using the blockchain technology concept of smart contracts, such as the access of land documents to a land inspector and the transfer of money from the buyer to the seller after successful land transfer verification. This system will resolve the issues of all three parties involved in the land registration process, eliminating the need for intermediaries such as real estate brokers. This system improves the land registration process while decreasing the possibility of fraud. The system also allows for land validation because immutable transactions are stored in the public ledger.

The [2] article outlines a ten-step decision-making process for determining whether blockchain is a good fit for LAS and what type of blockchain should be used. The paper uses the decision route to decide to implement the solution on a permissionless public blockchain. Anyone can join a permissionless public blockchain network and participate in transaction registration.

Regarding the rental housing market, there are many landlords, arbitrary fees, false rental information, etc., and it is difficult for the government to grasp and monitor the rental situation of rental housing. The [6] paper builds a residential rental alliance chain based on cryptographic algorithm blockchain technology. A smart contract is used by the landlord and tenant to sign the lease, establish the tenant, automatically pay and collect the rent on a regular basis, and return the tenant's rights when they are due. The alliance chain network's design incorporates an authentication department node and a record management node to validate the authenticity of housing source and personal identification information, secure rental information, register flat rental status, and realise secure lease effects. This method cuts out the middlemen, lowers costs, clarifies flat rental information, and aids government market oversight.

D. Tokenization

A token denotes a particular asset or function. Tokenization is the process of converting something valuable into a digital token that can be used on a blockchain platform. There are two ways to tokenize blockchain assets. They can represent both tangible assets like gold, real estate, and art as well as intangible assets like voting rights, property rights, and content licences. Almost anything can be tokenized as long as it is considered an asset that can be owned, has monetary value, and can be integrated into a larger asset market. Tokenization is an

old concept that predates blockchain technology. Tokenization has been used in the financial services industry to protect customers' personal information since the 1970s. Traditionally, sensitive data such as credit card numbers, Social Security numbers, and other personal information were converted into an alphanumeric string, which was then processed by a cryptographic algorithm to generate a unique token.

The article [4] goes over the legal, technical, and organizational aspects of real estate tokenization. It investigates the concept of technical neutrality, in which the old land registry is not necessarily replaced by blockchains, but people are given the option. The use of blockchain technology based on tokenization is explained in greater detail in this paper. Tokens are records on the blockchain that represent property and title rights. A token is a type of account unit that is linked to a user's address. Because of their private key, the user has complete control over the address. The token is linked to cadastral data (geodata) and property rights such as leases, mortgages, land, and other encumbrances and liens via technology. Blockchain records created by trusted third parties who can confirm ownership, deeds, and other transactions involving property rights ensure the link between property records and real estate and property rights. This is the first study to try to move an entire field of property rights and public records to Blockchain.

III. METHODOLOGY

This project aimed to create a web-based platform, which provides a secure and transparent platform for real estate investments using blockchain technology.

A. Planning Phase

During the planning phase, several steps were taken to ensure the success of the project. We began with a needs assessment to define the project goals and requirements. After determining the project scope and key deliverables, we conducted a market analysis to identify similar platforms, competitors, and market opportunities. Based on this data, we created a detailed project plan that included a timeline, budget, and resource allocation. This phase allowed us to lay the groundwork for the project and ensure we had a firm grasp of the goals, scope, and requirements before proceeding. [12]

B. Development Phase

We designed and developed the platform during the development phase in accordance with the project plan and requirements [12]. We began by defining the data entities and relationships and creating the database schema. The platform user interface and wireframes were then created. We used React JS for the front-end and its back-end features. To ensure secure and transparent investments, we integrated the Ethereum blockchain. Finally, we assessed the platform's functionality, security, and usability.

C. Main Components and Digital Assets

- **User interface (UI):** The user interface (UI) of Estate-O-Blockchain is intended to be intuitive and user-friendly, with clear navigation and simple information about real estate and investments. The platform's goal is to make real estate investing more accessible to a broader range of investors, and the UI interface plays a key role in achieving this goal by providing a seamless and efficient user experience.
- **Database:** The database is the central storage location for all platform-related data. Its purpose is to store information about properties, investors, investments, and transactions in a secure and efficient manner.
- **Blockchain integration:** The platform will be integrated with a blockchain, such as Ethereum, to provide secure and transparent real estate investments. The blockchain would be used to develop smart contracts that define investment terms such as token value, property details, and investor payouts.
- **Ethereum Smart Contract:** Smart contracts are self-executing computer programmes that run on a blockchain network to automate the contract verification, execution, and enforcement processes. Smart contracts are a necessary component of a blockchain-based real estate investment platform that enables the secure and transparent management of real estate fractional ownership. Smart contracts give investors greater control over their investments and eliminate the need for middlemen, making real estate investing more accessible and efficient to a broader range of investors.
- **Meta-wallet Connection:** A meta-wallet is a type of digital wallet that allows users to store and manage blockchain-based assets such as cryptocurrencies and tokens. It is also known as a Web3 wallet or an Ethereum wallet. By connecting a meta-wallet to the platform, users can manage their tokens and other Ethereum-based assets. Furthermore, Estate-O-Blockchain offers a more integrated investment ecosystem that enables token management and trading through a well-known and trusted wallet platform.
- **Tokenization:** Tokenization is the process of converting ownership of property into digital tokens that can be traded on a blockchain network [10]. Tokenization enables fractional ownership, allowing multiple investors to own a portion of a property instead of the entire property. Each token represents a unique aspect of the asset, such as a percentage of the property's value or a specific amount of rental income. Tokenization is an important feature of the Estate-O blockchain that allows for partial ownership while improving the liquidity, transparency, accessibility, and efficiency of real estate investments [11].

IV. IMPLEMENTATION

A. Architecture Components

- **React:** React is a well-known open-source JavaScript library for building user interfaces. It gives developers the ability to create and manage modular, reusable components. To improve performance and user experience, React uses a virtual DOM (Document Object Model) to update the user interface without requiring a page refresh. It also supports server-side rendering, which simplifies the development of universal web applications that can run on both the client and the server. To create complex and scalable web applications, React is frequently used in conjunction with other tools and libraries such as Redux, Webpack, and Babel. When developing a blockchain website, React can be used to create the front-end of a web application that interacts with the blockchain.
- **Remix:** Remix is a popular web-based integrated development environment (IDE) for Ethereum smart contracts. It provides a simple interface for developers to write, compile, and deploy Ethereum contracts on the blockchain. Remix also includes a set of debugging and testing tools to help developers identify and fix bugs in smart contracts before they are deployed to the live network. Remix allows developers to create, test, and deploy smart contracts in a simple and efficient manner.
- **Web3:** Web3 refers to the third generation of web technologies based on blockchain networks. It enables developers to build decentralised applications (dApps) that interact with blockchain networks like Ethereum to execute smart contracts, manage digital assets, and perform other tasks. Web3 is supported by a number of tools and protocols, including the Ethereum Virtual Machine (EVM), the Solidity programming language, and the Web3.js JavaScript library. Overall, Web3 offers a new paradigm for developing decentralised, trustless, and secure applications, with the potential to transform a wide range of industries and business models [9].
- **IPFS:** The InterPlanetary File System (IPFS) is a protocol and network that enables decentralised and distributed file storage and sharing. IPFS stores and distributes files through a peer-to-peer network of nodes rather than a centralised server or a single point of failure. IPFS accomplishes this by assigning each file a unique hash value that can be used to retrieve the file from any node on the network. Versioning, deduplication, and content-based addressing are also included in IPFS, making it a robust and adaptable solution for decentralised file storage and distribution. Additionally, IPFS can be used in conjunction with blockchain networks to enable decentralised applications (dApps) that require decentralised data and asset storage [8].
- **Firestore:** Firestore from Google Cloud Platform is a NoSQL database that allows developers to store and manage data for their applications. Firestore can be used to store information about properties, investors,

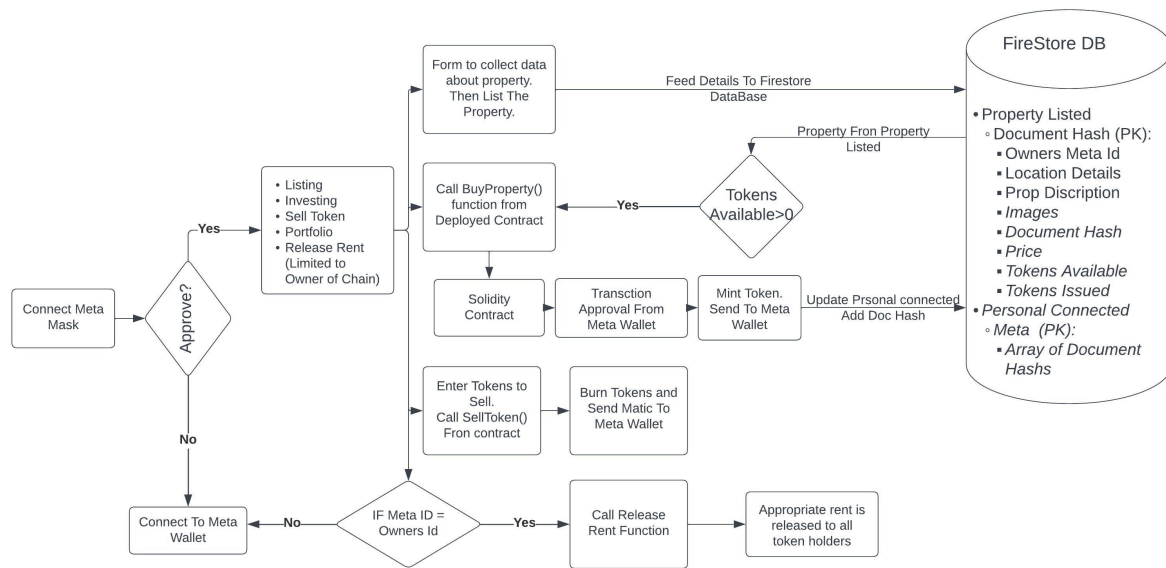


Fig. 1. Architecture Diagram

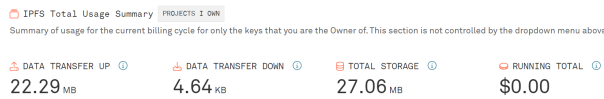


Fig. 2. IPFS Summary

transactions, and other topics. This information can be easily retrieved and edited using Firestore's powerful query and indexing capabilities. Furthermore, Firestore includes security rules that enable developers to control database access and ensure that only authorised users can read or write data. The EOB platform can provide investors with a seamless and secure experience for buying, selling, and managing fractional ownership of real estate by integrating Firestore with other technologies such as React, Ethereum blockchain, and IPFS.

- **Solidity:** Solidity is a high-level programming language that is used to build smart contracts on blockchain platforms such as Ethereum. It is a contract-based, object-oriented programming language that allows developers to create programmes that interact with the blockchain. Solidity provides data types, control structures, functions, and libraries to help you write smart contracts. It also includes security features that aid in the prevention of common flaws such as re-entry attacks and integer overflow/underflow. The Solidity code is compiled into bytecode that can be executed by the Ethereum Virtual Machine (EVM), a secure and transparent runtime environment for blockchain smart contracts.
- **Polygon:** Polygon, formerly known as Matic Network, is an Ethereum-based Layer 2 scaling solution designed to improve the scalability and usability of decentralised applications. It makes use of sidechains and the Plasma

framework to make transactions faster and cheaper. Polygon is Ethereum-compatible and supports Solidity-based smart contracts, making it an excellent choice for EOB-like projects. Polygon can help EOB provide faster transaction times, lower transaction fees, and a better overall user experience. Polygon also offers a wide range of developer tools and resources, including its own Web3 provider, to help developers integrate their projects into the Polygon network.

This study presents the development of a decentralized application using ReactJS as a frontend to enable property listing on the blockchain. Firestore is used to store property details and images, while IPFS is used for storing property documents with a unique hash serving as the Firestore database key. The application enables users to purchase property tokens by invoking the buyProperty function in the deployed contract and paying through Metamask. Upon receiving the required token value, a new token is minted and transferred to the user's Meta wallet. Similarly, the sellToken function can be used by users to sell their tokens and receive appropriate payment. The contract is written in Solidity, a programming language designed specifically for smart contracts on the Ethereum blockchain. The use of Solidity provides security and trustlessness, essential features for property transactions. The findings demonstrate the potential of blockchain technology for secure and efficient property transactions, indicating a promising avenue for future research.

B. Events and Functions

The blockchain is built on logs that contain records of events, functions, and their arguments written by hosted smart contracts. The Ethereum blockchain hosts the following smart contracts:

- BuyToken- When the buyer completes the transaction, a token is generated.

```
function buyProperty(uint256 _tokenAmount) public payable {
    uint256 toPay = _tokenAmount * msg.value;
    // require(msg.value >= toPay, "Less ether sent");

    tokensBought += _tokenAmount;
    holderCount++;
    holders[holderCount] = msg.sender;
    feesCollected += toPay*platformFee/100;
    _mint(msg.sender, _tokenAmount);
}
```

Fig. 3. Buy Token Smart contract

- SellToken- When a user sells a token, the appropriate amount is transferred to the meta wallet and the token is burned.

```
function sellToken(uint256 _tokenAmount, uint256 _price) public payable {
    _burn(msg.sender, _tokenAmount);
    payable(msg.sender).transfer(_tokenAmount*_price);
}
```

Fig. 4. sell Token Smart contract

- Release Rent- When the house owner pays the rent on the interest, the money is automatically distributed to all token holders.

```
function releaseRent() public onlyOwner {
    uint256 amountPerToken = (tokensBought*25)/120000;
    for(uint256 i = 1; i < holderCount+1; i++) {
        payable(holders[i]).transfer((balanceOf(holders[i]) * amountPerToken));
    }
}
```

Fig. 5. Release Rent Smart contract

- PayOwner- When someone purchases the token, the corresponding amount is sent to the property owner.

```
function payOwner(uint256 _amount, address _account) public payable onlyOwner {
    payable(_account).transfer(_amount*80/100);
}

function withdrawFees() public onlyOwner {
    payable(msg.sender).transfer(feesCollected);
}
```

Fig. 6. PayOwner Smart contract

V. RESULTS

A. Landing-page

The landing page for EOB is purposefully designed to attract visitors and encourage them to take specific actions. Users are introduced to the EOB platform, which uses blockchain technology for real estate investment, upon landing on the page. Visitors must first connect their Metamask account, a digital wallet commonly used for blockchain transactions, before proceeding.

Users gain access to all features of the EOB website by connecting to their Metamask account. They can then view their portfolio, track their investments, and manage their token holdings through their account dashboard. Users can also search for properties on the platform, view property details, and assess investment opportunities.

Users can purchase tokens representing fractional ownership of certain properties through the EOB platform. These tokens can be bought and sold on the platform, providing users with liquidity and flexibility in their investments. Through the innovative use of blockchain technology, it provides a seamless and user-friendly experience for individuals interested in real estate investing.

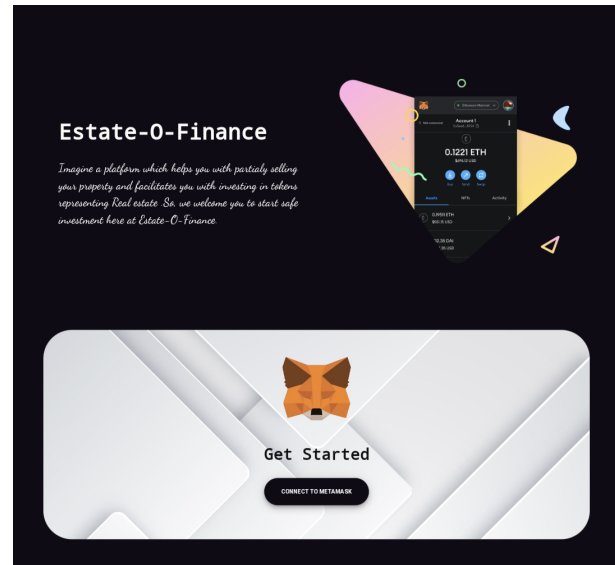


Fig. 7. landing page

B. Listing

This component allows people to list their properties on the created website. To list a property on the website, you must provide detailed information about the property, including the address, type, size, number of bedrooms and bathrooms, and any special features or amenities. You will also need to provide high-quality photos of the property to showcase it to potential investors.

In order to determine the property's market value and potential return to investors, you will also need to provide a professional appraisal or valuation. All legal documents related to the property, such as title deeds, mortgage documents, and building permits, must also be provided.

Finally, you must provide contact information for the property's owner or manager, including name, email address, and phone number. Overall, the goal is to provide enough information about the property and investment opportunity for potential investors to make informed decisions and invest confidently.

When a user submits data to the application, the data would be temporarily stored in memory or a local database. The application would then use IPFS to store the data in a decentralized network. The IPFS network would generate a unique hash that identifies the data, and this hash is stored in a traditional database.

Fig. 8. Listing

estate-o-finance

+

Start collection

PersonalList

PropertiesListed

+

Add document

QmB3TfYodswYFafY4J1TU14x3Upjg

QmW7D3Q4Z1pCvEJh13U1PDjREFB9

QmX7UDCy7QvRHE15qndA55eJNgst

QmavvHvGehUCnRd2x11B015FRgB

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Address: "402/ Neeminath apt, Khandwala lane, Malad East"

Description: "Good Locality And Fully Furnished Home"

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MetaId: "0a7ac08129e8d6f6bea8075137ea2c0ffed557d8"

OwnerEmail: "Mumbai"

PinCode: "400097"

Fig. 9. Listed Property in database

C. Portfolio

The "Portfolio" page on the EOB website serves as a central hub where users can view all of the properties they have listed on the platform. When a user requests data associated with a specific property, the Portfolio application employs a unique hash assigned to that property. This hash serves as a reference point for the application, allowing it to retrieve the corresponding data from the InterPlanetary File System.

The Portfolio application ensures that property data is securely stored and easily accessible by leveraging IPFS. When the data is retrieved from the IPFS, it is presented to the user in a user-friendly format, containing vital information about the property such as its location, details, ownership status, rental income, and other pertinent information.

D. Invest and Sell

The "Invest" page displays all listed properties in which an investor may invest. This page contains important information for investors who wish to purchase a token. The site provides details about the property, including location, size, and special features, as well as photos, videos, and floor plans. Important information about the token for sale, such as the token price and the total number of tokens available.

Transaction details, legal and regulatory information, and disclaimers are also provided to help investors make informed decisions. Overall, the investment page provides comprehensive information and resources for investors interested in purchasing real estate tokens.

On the "Invest" page, investors can also buy tokens from potential properties. Once the user buys the token, a smart contract -" BuyProp " is executed, which creates a token and sends it to the investor's meta-wallet.

On the "Sell" page, investors can sell the tokens they currently own. Once the user sells the tokens, a smart contract

-" SellProp " is executed, which burns the sold token and sends the corresponding amount to the seller in the meta-wallet.

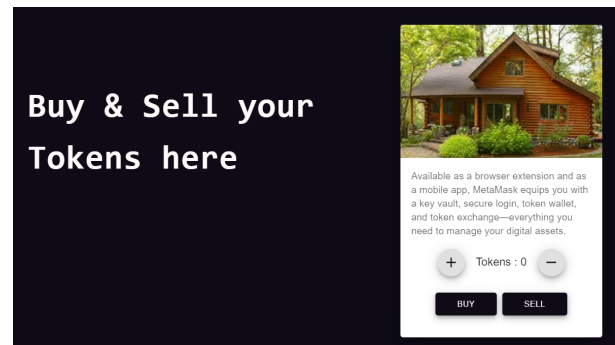


Fig. 10. Buy and Sell

VI. CONCLUSION

Estate-O-Blockchain can revolutionize the real estate investment industry through blockchain technology, tokenization, and smart contracts. Digitizing real estate assets into blockchain-based tokens can enable fractional ownership and make real estate investing accessible to a broader audience. Tokenization can also increase liquidity and transparency, streamline the investment process, and reduce the need for intermediaries.

In addition, smart contracts can automate many of the processes associated with real estate investing, such as KYC/AML verification, rent distributions, and property management, an easy way out from hefty taxes, liquidity in the market, and equal opportunities. This increases efficiency and lowers costs, making real estate investment more attractive to investors. Meta-wallet integration can create a more integrated investment ecosystem, allowing users to manage their investments and other Ethereum-based assets through a single dashboard. This provides a seamless and secure way to manage investments and optimize portfolios.

This project has the potential to transform the traditional real estate investment industry by making it more accessible, efficient, and transparent. As blockchain technology becomes more prevalent, more real estate investment platforms are expected to emerge, providing investors with a variety of investment options and the potential for significant returns.

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